PORTFOLIO PROJECT DOCUMENTATION

SME Loan Pre-Screener

Explainable, Fair AI Loan Pre-Screening for South African SMEs

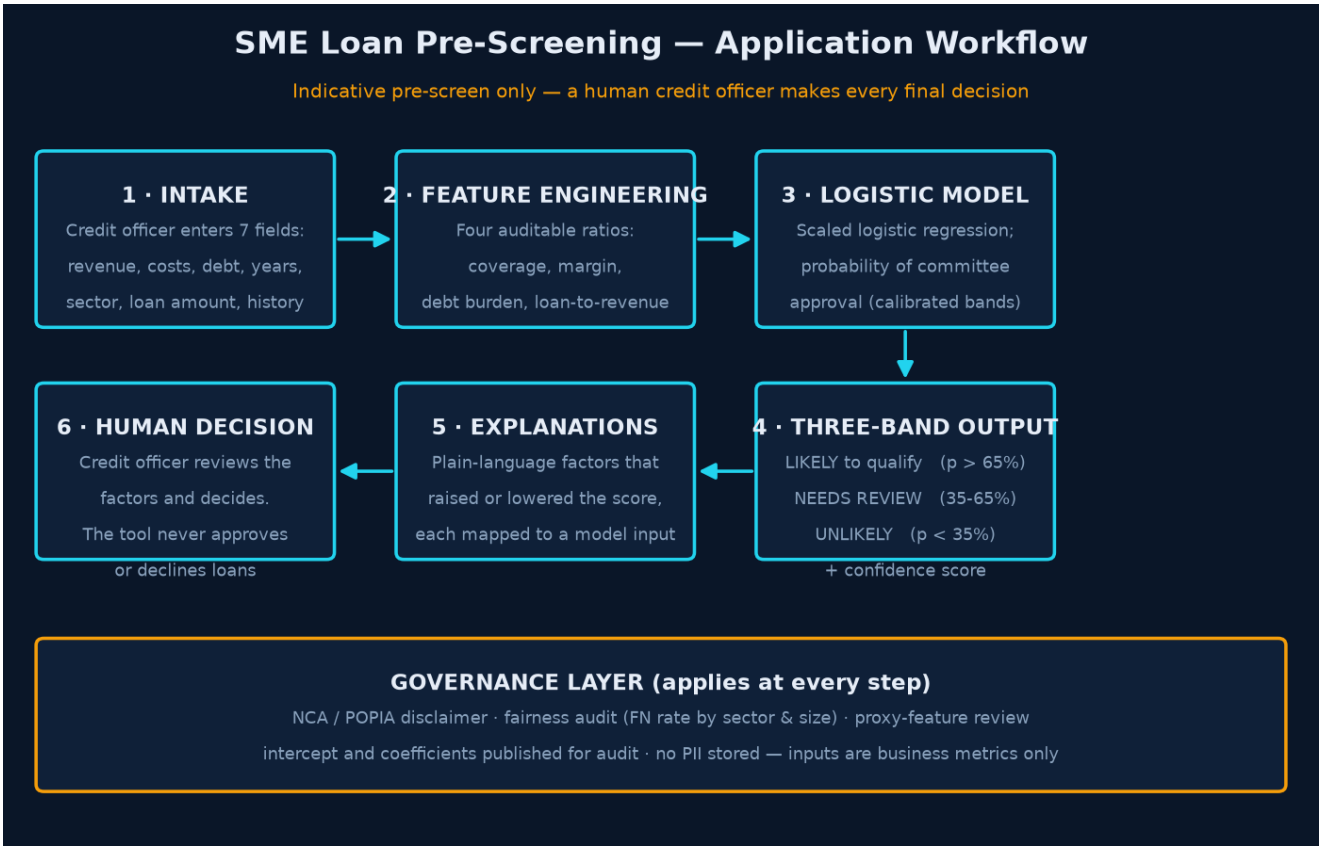
Tshepiso Freddy Thosago · October 2026 · Synthetic data only · Indicative tool, human judgment final

1. Problem statement

South African SMEs seeking working capital face slow, opaque credit processes. This prototype is an **AI co-pilot for credit officers**: it accelerates evaluation with a three-band indicative assessment and plain-language reasons, and it never approves or declines a loan itself.

LIKELY TO QUALIFY **NEEDS REVIEW** **UNLIKELY TO QUALIFY**

2. Architecture



> Seven lightweight inputs are engineered into affordability ratios, scored by an interpretable logistic model, mapped to calibrated probability bands, and explained with plain-language factors. A human credit officer makes the final call at every outcome.

3. Data foundation

- **Source:** fully synthetic, seeded generator (`data/generate_dataset.py`), 800 applications; no real customer data.
- **Target engineering:** a simulated credit committee approves an application when an affordability score (instalment coverage, margin, debt ceiling at 30%, trading years, prior history, sector premium) clears a calibrated logistic threshold plus noise. Approval rate: **33.3%**.
- **Missing values:** debt-free applicants (no existing repayments) are explicitly imputed to 0 and documented; no rows dropped.

4. Model selection and evaluation

Model	Precision	Recall	F1	ROC-AUC	Confusion (TN/FP/FN/TP)
Rule-based baseline	0.625	0.379	0.472	0.000	119/15/41/25
Logistic regression (deployed)	0.494	0.636	0.556	0.722	91/43/24/42
Gradient boosting (comparison)	0.523	0.348	0.418	0.675	113/21/43/23

Held-out test set: 200 applications (25% stratified split, seed 42). The logistic model is deployed because it matches the gradient booster's discrimination while remaining fully auditable. `class_weight='balanced'` deliberately trades precision for recall: **false negatives harm the applicant** (a fundable SME is discouraged), while false positives cost the lender underwriting time and are caught in detailed review.

False-positive vs false-negative cost

	Meaning in this prototype	Who bears the cost
False positives (43)	SME pre-screened as LIKELY to qualify but the committee would decline.	The lender
False negatives (24)	Fundable SME pre-screened as unlikely/negative but the committee would approve.	The applicant

5. Explainability (non-negotiable)

Every assessment lists the top factors that raised or lowered the score, in credit-officer language. Factors are computed from the same audited ratios the model sees, for example:

- Free cash of R67,000 per month covers the estimated new instalment of R16,667 about 4.0 times.
- Existing debt repayments consume 40% of monthly revenue, above the 30% affordability ceiling.
- The requested R600,000 is about 6.7 months of revenue - a large ask relative to turnover.

6. Fairness audit

False-negative rate among approvable applications, by sector (test set):

Sector	N	False negatives	FN rate
Construction	32	6	0.67
Manufacturing	23	7	0.50
Retail	49	8	0.42
Hospitality	26	1	0.25
Services	70	2	0.10

By business size (revenue median split): smaller half **0.26**, larger half **0.50**.

> **Disclosure:** the sector gaps are engineered into the synthetic generator (construction and hospitality carry a committee risk premium) so the audit has a real disparity to surface. In production the monitoring pipeline is the deliverable: FN-rate gaps by sector and size would be tracked monthly, and persistent gaps trigger model review. No protected characteristics are inputs; the small/large revenue split is monitored as a proxy-risk feature.

7. Responsible AI and regulatory framing

- **Indicative only:** prominent, persistent disclaimer - the tool never approves or declines.
- **NCA and POPIA:** production deployment would require National Credit Act registration and POPIA compliance; out of scope for this prototype and stated on the interface.
- **Proxy features:** sector and size are retained for explainability but flagged and audited; protected characteristics are excluded by design.

8. Reproduce

```
git clone https://github.com/tshepisofrominnostation/sme-loan-prescreener
pip install -r requirements.txt
python data/generate_dataset.py && python -m src.train
streamlit run app.py
```

Web prototype: **https://tshepisofrominnostation.github.io/sme-loan-prescreener/** - the same model runs client-side with exact probability parity against Python.

9. Limitations and future work

- Synthetic labels approximate, not replace, real credit committee behaviour.
- Binary committee target collapses a spectrum; three bands recover the middle ground for human review.
- Future: calibration validation curves, PR-AUC threshold optimisation, SHAP factor attribution as a secondary explainer, and lender-side A/B evaluation.